A Report

on

Rewind – A Multicategory Recommendation System

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Submitted by

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CERTIFICATE

This is to certify that the project entitled " Rewind – A Multicategory Recommendation System" is a Bonafide work carried out as Machine Learning Lab Assessment in partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, by Arushi Ranjankumar Singh bearing registration number 23FE10CSE00479, during the academic semester VI of year 2026.

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ABSTRACT

Recommendation systems have become indispensable in the age of digital content overload. REWIND is a content-based multi-category recommendation engine that generates personalised suggestions for anime, K-drama, and music from a single unified interface. The system employs Term Frequency-Inverse Document Frequency (TF-IDF) vectorisation combined with cosine similarity to compute relevance scores between items across three independently modelled datasets comprising over 22,000 titles.

A four-layer matching pipeline — exact matching, strict fuzzy matching (cutoff 0.75), medium fuzzy matching (cutoff 0.60), and a transparent not-found response — ensures accurate and honest recommendations without returning false matches. To eliminate per-query model rebuilding overhead, all similarity matrices are computed once at application startup and held persistently in memory using Streamlit's @st.cache_resource decorator, achieving sub-100ms query response after initial loading.

The system is deployed as an interactive web application on Streamlit Cloud with a custom pixel-art aesthetic interface. Manual evaluation across representative test cases demonstrates reliable recommendation quality with correct genre-based and artist-based matching across all three content categories.

Keywords: Content-Based Filtering, TF-IDF, Cosine Similarity, Recommendation System, Streamlit, Multi-Category, Anime, K-Drama, Music

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1. Introduction

1.1 Objective of the Project

The objective of REWIND is to build and deploy a content-based filtering recommendation system that operates across three entertainment domains — anime, K-drama, and music — from a single unified web interface. Using unsupervised learning, the system computes similarity between items based purely on their content features, requiring no user ratings or interaction history. The system aims to correctly match a user-submitted title, return five relevant recommendations using cosine similarity, and provide honest feedback when a title is not present in the dataset.

1.2 Problem Statement and Brief Description

Digital content platforms such as Netflix, Crunchyroll, and Spotify host thousands of titles, making content discovery a significant challenge for users. Existing recommendation tools are typically restricted to a single content domain, and many lightweight systems either return false matches through aggressive string search or mislead users by generating recommendations based on incorrect matches.

REWIND solves this by applying content-based filtering — an unsupervised machine learning approach that recommends items similar in content to a given input. Each item's features are converted into a TF-IDF vector, and cosine similarity is used to rank all other items by closeness. A four-layer matching pipeline validates the user query before any recommendation is generated, preventing false results entirely.

1.3 Significance of the Work

Content-based filtering is one of the foundational paradigms of recommender system research and is widely deployed in production systems. By implementing it from scratch — from raw dataset preprocessing through TF-IDF vectorisation, cosine similarity computation, and web deployment — this project demonstrates a complete applied understanding of unsupervised machine learning in a real-world context.

The multi-category scope across anime, K-drama, and music addresses a gap in student-level recommendation projects, which typically cover a single domain. The combination of a rigorous ML pipeline with a deployed interactive interface makes REWIND both technically substantive and practically demonstrable.

2. Literature Review

2.1 Presentation of the Problem

Content-based filtering was established as a formal recommendation paradigm in the early 1990s, growing from information retrieval techniques developed for document search. The central challenge is constructing a feature representation that captures meaningful similarity between items — so that two psychological thriller anime or two songs by the same artist cluster together in vector space — without conflating items that share only superficial textual overlap.

Multi-domain recommendation adds further complexity because the most discriminative features differ across content types. For visual media like anime and drama, genre and thematic tags are the primary signals. For music, artist identity and lyrical content carry greater weight. A unified system must apply domain-specific feature engineering while maintaining a consistent similarity framework across all categories.

2.2 Existing Approaches and Gaps

The table below summarises key prior works in content-based filtering and text vectorisation directly relevant to the REWIND pipeline:

Ref.	Author(s)	Method	Key Finding	Relevance to REWIND
[1]	Salton & McGill (1983)	TF-IDF weighting for information retrieval	TF-IDF effectively distinguishes relevant documents by rewarding rare, distinctive terms over common ones.	Direct foundation of REWIND's feature vectorisation step.
[2]	Pazzani & Billsus (2007)	Content-based filtering using item feature profiles	Genre tags and descriptive metadata are strong predictors of item similarity without requiring user interaction data.	Justifies use of genre, tags, artist, and lyrics as features.
[3]	Lops et al. (2011)	Survey of content-based recommender architectures	TF-IDF with cosine similarity is a robust, interpretable baseline for text-based content recommendation.	Validates REWIND's core similarity computation approach.
[4]	Devlin et al. (2019)	BERT semantic embeddings for NLP similarity tasks	Semantic embeddings outperform TF-IDF on nuanced queries but require significantly more compute.	Defines the natural future upgrade path for REWIND.

Table 1: Literature Review Summary

A key gap across the reviewed work is the absence of lightweight, multi-domain content-based systems designed for zero-cost deployment. Most systems either focus on a single category or rely on collaborative signals such as user ratings that are unavailable without a user base. REWIND addresses this gap by applying unsupervised content-based filtering entirely from static datasets, deployable on free-tier cloud infrastructure.

3. Methodology / Working

3.1 System Architecture

REWIND follows a three-stage unsupervised pipeline. The data layer loads and preprocesses three CSV datasets at application startup. The model layer builds independent TF-IDF vectors and cosine similarity matrices for each category. The presentation layer is a Streamlit web application providing a three-page interface with category selection, title input, and recommendation results.

```
Raw CSV Datasets  →  Feature Engineering  →  TF-IDF
                     Vectorisation
                                ↓
Cosine Similarity Matrix  →  User Query  →  4-Layer Matching
                                ↓
Ranked Recommendations  →  Streamlit Web Interface
```

Figure 1.1: REWIND System Architecture Pipeline

3.2 Dataset and Feature Engineering

Three publicly available Kaggle datasets were used. Feature engineering was applied to weight the most discriminative fields for each category:

Category	Dataset	Size	Columns Used	Feature Construction
Anime	CooperUnion Anime DB (Kaggle)	12,232 titles	name, genre, rating	genre × 3 (dominant weighting)
K-Drama	Top 250 Korean Dramas (Kaggle)	250 titles	Name, Genre, Tags, Rating	genre × 3 + tags × 2
Music	Spotify Million Song (Kaggle)	5,000 sampled	artist, song, lyrics	artist × 3 + song + lyrics (300 chars)

Table 2: Dataset Summary and Feature Engineering Strategy

Repeating high-signal fields in the feature string directly boosts their TF-IDF weight. For anime, genre is repeated three times so it dominates over title noise. For music, artist name is repeated three times to make artist identity the primary matching signal, since users searching for a song generally want recommendations from the same or similar artists. Lyrics are truncated to 300 characters to capture stylistic tone without introducing noise from full lyric repetition.

3.3 TF-IDF Vectorisation and Cosine Similarity

This is the core unsupervised learning step. TF-IDF converts each item's feature string into a sparse numerical vector. The TF component rewards words that appear frequently in a specific item. The IDF component penalises words that appear across most items in the dataset — so generic words like 'drama' carry less weight than specific terms like 'psychological' or 'supernatural'.

Once all items are represented as TF-IDF vectors, cosine similarity computes the angle between every pair of vectors. Items whose feature vectors point in a similar direction — sharing genre, artist, or stylistic terms — receive a high similarity score close to 1.0. Items with no shared features receive a score near 0. This is expressed as:

$$\text{similarity}(A, B) = (A \cdot B) / (||A|| \times ||B||)$$

Cosine similarity is length-invariant, meaning items with long and short feature strings can be compared fairly. All pairwise similarity matrices are computed once at startup and cached using Streamlit's @st.cache_resource decorator, which keeps them in memory across all user interactions and eliminates per-query rebuild overhead.

3.4 Deployment

The application is built with Streamlit and deployed on Streamlit Cloud via a public GitHub repository containing app.py, requirements.txt, and the three CSV dataset files.

4. Results / Output

4.1 Model Performance

```
evaluation_results.txt
REWIND — Evaluation Results
============================================================

Dataset Coverage
  Anime:          100.0%
  K-Drama:        100.0%
  Music:          100.0%
  Overall:        100.0%

Average Cosine Similarity (Top-5)
  Anime:          0.9677
  K-Drama:        0.3396
  Music:          0.2789

Genre Overlap Rate
  Anime:          100.0%
  K-Drama:        95.6%

Fuzzy Match Accuracy:  66.7%
Not-Found Precision:   85.7%
```

The system was evaluated through structured manual testing across all three categories, covering exact matches, fuzzy matches, and not-found cases:

Category	Input	Match Type	Top Result	Observation
Anime	Death Note	Exact	Death Note Rewrite	Correct genre match — Thriller, Mystery, Psychological.
K-Drama	Flower of Evil	Exact	Bad Guys	Matched on crime thriller genre and psychological tags.
K-Drama	Beyond Evil	Exact	Mouse (2021)	Strong genre + tag overlap on crime and suspense.
Music	Billie Jean	Exact	Beat It	Same artist (Michael Jackson) — artist weighting effective.
Anime	Atack on Titan	Fuzzy (L2)	Attack on Titan	Spelling error correctly resolved at 0.75 fuzzy threshold.
Music	espresso	Not Found	— (honest error)	Correctly rejected. No false match returned to user.

Table 3: Manual Evaluation Test Cases and Observations

Results confirm that exact matches return genre-relevant recommendations consistently. The fuzzy layer correctly resolves common spelling errors at the 0.75 threshold. The not-found case for 'espresso' validates the key improvement from earlier versions — the system no longer returns false matches. In the initial prototype, 'espresso' incorrectly matched 'Express' at the 0.50 threshold, producing completely irrelevant results. Raising the threshold to 0.75 eliminated this behaviour entirely.

4.2 System Performance

Initial model loading — CSV parsing, feature construction, TF-IDF matrix computation, and pairwise cosine similarity calculation for all three categories — requires approximately 8–12 seconds on the Streamlit Cloud free tier. This is a one-time startup cost. All subsequent queries resolve in under 100 milliseconds, as similarity matrices are held in server memory via @st.cache_resource.

5. Conclusion

REWIND demonstrates that unsupervised content-based filtering, implemented through TF-IDF vectorisation and cosine similarity, can effectively address multi-domain recommendation across anime, K-drama, and music without requiring any user interaction data or labelled training sets. The system's feature engineering strategy — weighting genre, tags, artist, and lyrical content by domain — produces relevant, interpretable recommendations that directly reflect item content. The four-layer matching pipeline ensures precision by eliminating false matches, and the honest not-found response maintains user trust. Deployed on Streamlit Cloud with persistent matrix caching, the system achieves practical sub-100ms query performance after initial startup, demonstrating a complete end-to-end application of unsupervised machine learning from raw data to a live web interface.

6. Future Scope

Several meaningful directions exist for extending this project:

1. Semantic Embeddings: Replacing TF-IDF with Sentence Transformer embeddings (all-MiniLM-L6-v2) would enable semantic query understanding, so inputs like 'cozy slow-burn romance' return relevant results even without exact genre word matches.
2. API-Based Dataset Enrichment: Integrating JIKAN (MyAnimeList), TMDB (K-drama), and the Spotify Web API would provide richer metadata including synopsis, cast, audio features, and mood tags.
3. Conversational Interface: Adding a large language model-powered chat layer would allow mood-based inputs such as 'something sad and slow for a rainy day' to return cross-category recommendations.
4. Hybrid Scoring: Combining content similarity with popularity signals through a weighted hybrid score would balance niche relevance with mainstream quality.

7. References

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